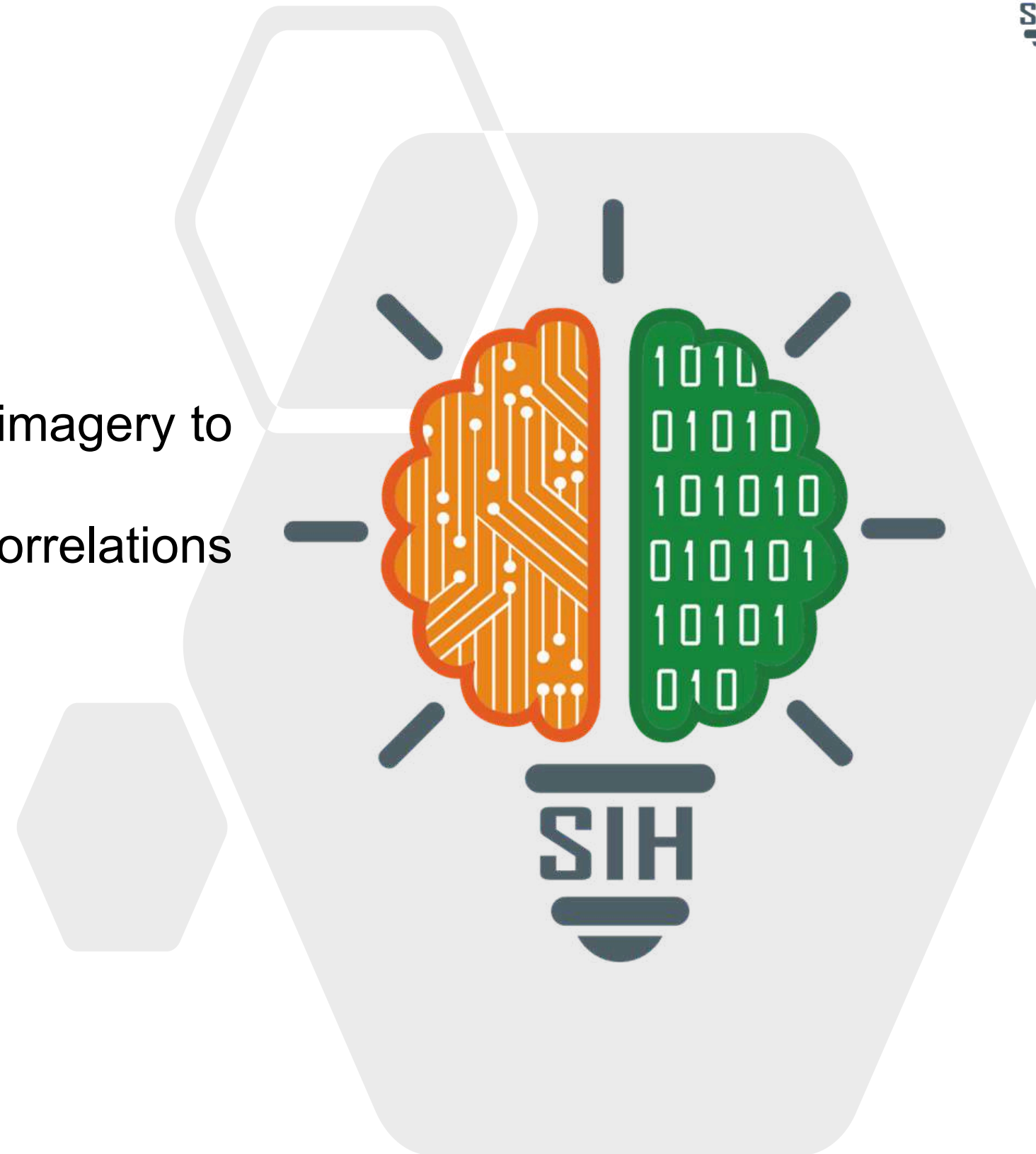


# SMART INDIA HACKATHON 2026



- **Problem Statement ID** – 26143
- **Problem Statement Title** - Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.
- **Theme** - Disaster Management
- **PS Category** - Software
- **Team ID** - IIC\_TMSL\_P8\_S12
- **Team Name** - Two Cmds & Three Ctrl's



# DEVICED SOLUTION



## Detailed explanation of the proposed solution

1

One SAR satellite image in → ranked vessel shortlist + signed PDF incident report out, in 24 seconds on a laptop.



2

**10 automated stages:**  
detect slick → screen out look-alikes → measure geometry → run drift backward to origin → release window → run it forward to forecast → reconstruct vessel traffic → filter irrelevant traffic → score and rank → dispatch report.

1

2

3

4

5

6

7

8

9

10

End-to-end pipeline (10 stages)

3

**Working prototype, not a concept:**  
6 screens, 776 automated tests passing, runs offline.

6 Screens

776 Tests Passing

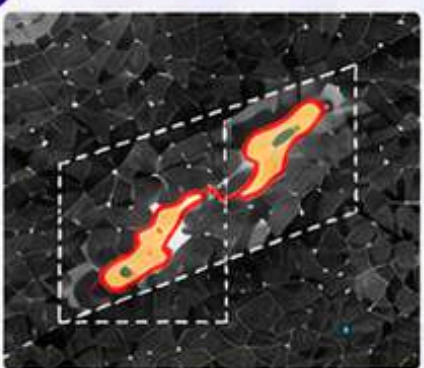
Runs Offline



## How it addresses the problem

a

**Detect, characterise, age**





IoU 0.769  
(on held-out patches)



9 regions, 223.19 km<sup>2</sup>  
measured on a sphere



Spill age with a resolvability test  
(not a midpoint)

b

**Ocean & met data, origin point & time, future flow**





Current 0.072 m/s  
+ wind drift 0.168 m/s



Origin 54.64°E, 25.52°N  
P90 radius 11.26 km



Release window 24 h



Forward forecast  
from same integrator

c

**Reconstruct traffic, filter the irrelevant, score it**

987 synthetic AIS reports  
(10 vessels)



2 relevant  
8 filtered out



Six weighted components (sum = 100)



Top candidate  
91.5



## Innovation and uniqueness

1

**A classical baseline we built to compete against ourselves**



Dark-spot threshold  
0.676 IoU



Our U-Net  
0.769 IoU



+0.093  
gain

2

**The spatio-temporal match is genuinely temporal**



Sentinel-1  
(SAR imagery)



ERA5 (wind)  
— reader built;  
synthetic in demo



CMEMS (current)  
— reader built;  
synthetic in demo



Synthetic AIS  
(MarineCadastre schema)



**AI + Physics Pipeline**  
• Segmentation  
• Drift Model  
• Track Analysis  
• Scoring



Incident Report  
(PDF)

3

**Every output states what it does not know**



24.0 s  
per scene



Run offline  
(on laptop)



776  
Tests passing

# TECHNICAL APPROACH





### Feasibility



#### Feasibility Analysis



**Runs on a consumer laptop, CPU only.**  
No GPU, no cloud spend, no licence fee.



**24.0 s per scene.**  
End-to-end processing time on a single machine.



**Inputs are free and operational.**  
Sentinel-1, ERA5, CMEMS, Synthetic AIS (MarineCadastré schema).



**Open data formats.**  
Uses MarineCadastré 17-column AIS schema for interoperability.



**Air-gapped deployment.**  
No internet required for core processing.



**Feasible with existing tools, data and infrastructure.**



#### Key Technical Capabilities



**SAR processing**  
(Sentinel-1 GeoTIFF)



**Ocean & met data**  
(ERA5 + CMEMS)



**Synthetic AIS**  
(MarineCadastré)



**Data integration**  
(17-column schema)



**Real-time pipeline**  
(24 s per scene)



### Viability



#### Challenges, Risks and Strategies

|    | Risk                                                                                                                                                                                                                                                                                                                                              | Strategy                                                                                                                                                                                                                                                                                |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 |  <p><b>Look-alikes</b><br/>Low wind, algal blooms, rain cells look like oil in SAR.</p>                                                                                                                                                                        |  <p><b>7-feature screen, all scale-invariant ratios.</b><br/>AUC 0.9573 in-domain; rejects 69.4% of dark regions on 2,290 published look-alike patches never trained on real gain, not solved.</p> |
| 02 |  <p><b>Splits grouped by parent acquisition.</b><br/>An earlier split leaked, we found it, fixed it, wrote the regression test and re-ran everything. Our margin over the classical baseline halved — +0.189 → +0.093. That second number is the real one.</p> |  <p><b>Split-by-parent acquisition handling.</b><br/>Leak fixed, regression test added, re-ran full pipeline. No more leakage, stable results.</p>                                                 |
| 03 |  <p><b>Licensed AIS unavailable</b><br/>to a student team.</p>                                                                                                                                                                                                 |  <p><b>Synthetic AIS in exact 17-column MarineCadastré schema.</b><br/>Diff-able against a real extract. Swapping in real feeds is a reader change, not a rewrite.</p>                             |
| 04 |  <p><b>Attribution has legal consequences.</b></p>                                                                                                                                                                                                             |  <p><b>A test asserts “guilty”, “culprit”, “responsible party” appear nowhere in the PDF — and that “priority candidate for investigation” does.</b></p>                                           |
| 05 |  <p><b>No Indian-water validation yet.</b></p>                                                                                                                                                                                                                 |  <p><b>0 of 1,200 scenes fall in 65°-95° E, 5°-25° N.</b><br/>One Sentinel-1 scene over the Gulf of Kutch closes it.</p>                                                                           |



#### Why It Is Viable



**Solves a real operational need.**  
Faster, objective vessel attribution and incident reporting for maritime authorities.



**Builds on existing, trusted data.**  
Open satellite and ocean data; standard AIS schema; no new infrastructure required.



**Scalable and repeatable.**  
Runs offline, on commodity hardware, with deterministic results.



**Long-term impact.**  
Supports cleaner oceans, safer navigation and better disaster response.

# IMPACT AND BENEFITS



## Environmental Impact



01

**Cleaner Oceans**  
Faster detection and accurate origin help contain spills and reduce long-term ecological damage.



Lower marine pollution



02

**Protects Marine Life**  
Limits harm to fish, coral reefs and coastal ecosystems by enabling quicker response.



Healthier ecosystems



03

**Preserves Biodiversity**  
Reduces contamination of critical habitats and supports healthier ocean ecosystems.



Stronger biodiversity



04

**Supports Climate Goals**  
Prevents long-lasting pollution and aligns with global sustainability commitments.



Lower carbon footprint



**Cleaner oceans. Healthier ecosystems.**  
**A safer, more sustainable planet.**



## Operational Benefits



01

**Real-time Response**  
Complete analysis and incident report in 24 seconds on a laptop.



24.0 s  
per scene



02

**High Accuracy**  
U-Net model achieves 0.769 IoU, with 9 regions and 223.19 km<sup>2</sup> measured on a sphere.



IoU 0.769  
(patch)



03

**Data-Driven Decisions**  
Combines SAR, ocean & met data, AIS and multi-layer analysis to rank vessels with confidence.



91.5%  
top candidate score

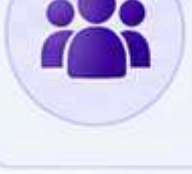


04

**Efficient & Scalable**  
Runs offline with open data sources and a simple, modular pipeline.



776  
tests passing



05

**Reduces Investigation Effort**  
Shortlists priority vessels and directs resources where they matter.



2 / 10  
relevant vessels identified



## Broader Impact



01

**Stronger Maritime Security**  
Improves situational awareness and supports agencies like the Coast Guard and NTRO.



Safer  
maritime borders



02

**Economic Protection**  
Prevents costly environmental damage, reduces cleanup costs and protects fisheries and coastal economies.



Lower  
response costs



03

**Better Policy and Planning**  
Rich, reliable data supports informed decision-making for marine management.



Evidence-based  
policies



04

**Sustainable Future**  
Builds a cleaner, safer and more resilient ocean for future generations.



Long-term  
resilience



**Safer coasts. Stronger economies.**  
**A more sustainable future.**

@SIH Idea submission- Template

<#>

# RESEARCH AND REFERENCES

## Datasets & Data Sources

### 1. Sentinel-1 SAR Oil Spill Image Dataset

- Trujillo-Acatitla et al. — IPICYT, Mexico
- Zenodo · CC BY 4.0
- DOI: 10.5281/zenodo.8346860
- Link: <https://doi.org/10.5281/zenodo.8346860>

### 2. DARTIS — SAR Dark-Formation Dataset

- Annotated dataset for oil-spill look-alike evaluation
- PANGAEA
- DOI: 10.1594/PANGAEA.980773
- Link: <https://doi.org/10.1594/PANGAEA.980773>

### 3. Copernicus Sentinel-1 GRD IW

- Satellite SAR imagery used by the pipeline
- Link: <https://dataspace.copernicus.eu/data-collections/copernicus-sentinel-missions/sentinel-1>
- Copernicus Data Space Ecosystem

### 4. CMEMS Global Ocean Physics

- Ocean-current forcing for particle advection and forecasting.

## Research & Methods

### 5. ERA5 Hourly Single Levels

- Wind / atmospheric forcing for drift modelling
- DOI: 10.24381/cds.adbb2d47
- Link: <https://doi.org/10.24381/cds.adbb2d47>
- Climate Data Store

### 6. MarineCadastre / NAIS AIS Data

- 17-column AIS schema used for vessel-trajectory reconstruction.
- Research & Methods

### 7. Marine Oil Spill Detection & Segmentation in SAR

- Marine Pollution Bulletin, 204, 116549 (2024)
- DOI: 10.1016/j.marpolbul.2024.116549
- Link: <https://doi.org/10.1016/j.marpolbul.2024.116549>

### 8. U-Net: Convolutional Networks for Biomedical Image Segmentation

- Ronneberger, Fischer & Brox — MICCAI 2015
- arXiv: 1505.04597
- Link: <https://arxiv.org/abs/1505.04597>